

ABHISHEK SENAPATI

Saw Swee Hock School of Public Health (SSHSPH)

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CURRENT POSITION

Senior Research Fellow

Saw Swee Hock School of Public Health (SSHSPH),

National University of Singapore, Singapore

EDUCATION

- **2021, Ph.D., Applied Mathematics, University of Calcutta, Kolkata, India (Research carried out at Indian Statistical Institute, Kolkata, India)**
Thesis Title: *The Spread and Control of Some Infectious Diseases in Patchy Environment - Model Based Studies*
- **2014, M.Sc., Applied Mathematics, University College of Science, technology and Agriculture, University of Calcutta, Kolkata, India**
- **2012, B.Sc., Mathematics, Presidency College, University of Calcutta, Kolkata, India**

RESEARCH INTERESTS

- Infectious disease modelling for public health preparedness and policy decisions
- Dengue and vector-borne disease epidemiology
- Vaccination and intervention impact assessment
- Pandemic preparedness and surveillance-informed modelling
- Climate-sensitive infectious disease risks

RESEARCH EXPERIENCES

- **Senior Research Fellow** (May 2026 - present) at SSHSPH, NUS, Singapore (Mathematical modeller in “**Dengue Modelling Consortium**” funded by **Wellcome Trust** and **Philanthropy Asia Alliance**)
- **Research Fellow** (May 2023 – April 2026) at SSHSPH, NUS, Singapore (Mathematical modeller in the project “**Mathematical modelling and health economic analysis of dengue vaccination strategies in Singapore**” funded by **NMRC, Singapore**)
- **Postdoctoral Researcher** (January 2021 – December 2022) at Center for Advanced Systems Understanding (CASUS), Goerlitz, Germany (Mathematical modeller in the project “**Where2Test**” funded by **SMWK, Germany**)
- **Senior Research Fellow** (funded by **CSIR, India**) (September 2017 – August 2020) at Indian Statistical Institute, Kolkata, India
- **Junior Research Fellow** (funded by **CSIR, India**) (September 2015 – August 2017) at Indian Statistical Institute, Kolkata, India

PUBLICATIONS

- **A. Senapati***, T-L. Thein, Y-S Leo, H. E. Clapham*. **(2026)**. “Adult dengue vaccination in a low transmission setting: A modelling study in Singapore”. ***PLoS Neglected Tropical Diseases***, 20: e0014062. doi:10.1371/journal.pntd.0014062.
- M. Davoodi, A. Batista, **A. Senapati**, W. Schlechte-Welnicz, B. Wagner, J. M. Calabrese. **(2026)**. “Optimal testing strategies in long-term care facilities during pandemics”. ***Journal of Optimization Theory and Applications***, 208: 1-33. doi:10.1007/s10957-025-02843-w.
- B. Maity, S. Banerjee, **A. Senapati**, J. Pitchford, J. Chattopadhyay. **(2025)**. “Coupling plankton and cholera dynamics: Insights into outbreak prediction and practical disease management”. ***PLoS Computational Biology***, 21: e1013523. doi:10.1371/journal.pcbi.1013523.
- A. Batista, **A. Senapati**, M. Davoodi, J. M. Calabrese. **(2024)**. “Personnel staffing and scheduling during disease outbreaks: A contact network-based analysis”. ***Computers & Industrial Engineering***, 193: 110112. doi:10.1016/j.cie.2024.110112.
- **A. Senapati***, A. Mertel, W. Schlechte-Welnicz, J. M. Calabrese. **(2024)**. “Estimating cross-border mobility from the difference in peak timing: A case study of Poland–Germany border regions”. ***Mathematics***, 12: 2065. doi:10.3390/math12132065.

- M. Davoodi, A. Batista, A. Mertel, **A. Senapati**, W. Abdussalam, J. Vyskocil, G. Barbieri, K. Fan, W. Schlechte-Welnicz, J. M. Calabrese. (2023). "A Web-Based COVID-19 Tool for Testing Residents in Retirement Homes: Development Study". *JMIR Formative Research*, 1: e45875. doi:10.2196/45875.
- M. Davoodi, A. Batista, **A. Senapati**, J. M. Calabrese. (2023). "Personnel Scheduling during the COVID-19 Pandemic: A Probabilistic Graph-Based Approach". *Healthcare*, 11: 1917. doi:10.3390/healthcare1113191.
- B. Maity, S. Banerjee, **A. Senapati**^{*}, J. Chattopadhyay. (2023). "Quantifying optimal resource allocation strategies for controlling epidemics". *Journal of the Royal Society Interface*, 20: 20230036. doi:10.1098/rsif.2023.0036.
- M. Davoodi, A. Batista, **A. Senapati**, A. Mertel, W. Schlechte-Welnicz, J. M. Calabrese. (2023). "On the optimal presence strategies for workplace during pandemics: A COVID-19 inspired probabilistic model". *PLoS One*, 18: e0285601. doi:10.1371/journal.pone.0285601.
- M. Roy, **A. Senapati**, S. Poria, A. Mishra, C. Hens. (2021). "Role of assortativity in predicting burst synchronization using echo state network". *Physical Review E*, 105: 064205. doi:10.1103/PhysRevE.105.064205.
- **A. Senapati**, S. Rana, T. Das, J. Chattopadhyay. (2021). "Impact of intervention on the spread of COVID-19 in India: A model based study". *Journal of Theoretical Biology*, 523: 110711. doi:10.1016/j.jtbi.2021.110711.
- S. Ghosh¹, **A. Senapati**¹, J. Chattopadhyay, C. Hens, D. Ghosh. (2021). "Optimal test-kit-based intervention strategy of epidemic spreading in heterogeneous complex networks". *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 31. doi:10.1063/5.0053262.
- S. Ghosh, **A. Senapati**, A. Mishra, J. Chattopadhyay, S. K. Dana, C. Hens, D. Ghosh. (2021). "Reservoir computing on epidemic spreading: A case study on COVID-19 cases". *Physical Review E*, 104: 014308. doi:10.1103/PhysRevE.104.014308.
- **A. Senapati**, P. Panday, S. Samanta, J. Chattopadhyay. (2020). "Disease control through removal of population using Z-control approach". *Physica A: Statistical Mechanics and its Applications*, 548: 1123846. doi:10.1016/j.physa.2019.123846.
- **A. Senapati**, S. Biswas, M. Martcheva, J. Chattopadhyay. (2019). "Bifurcation and optimal control of a Zika-Dengue co-infection model". *Book Chapter in Contemporary Research in Mathematical Biology*. doi:10.1142/12639.
- **A. Senapati**^{*}, T. Sardar, K. S. Ganguly, K. S. Ganguly, A. K. Chattopadhyay, J. Chattopadhyay. (2019). "Impact of adult mosquito control on dengue prevalence in a multi-patch setting: A case study in Kolkata (2014–2015)". *Journal of Theoretical Biology*, 478: 139-152. doi:10.1016/j.jtbi.2019.06.021.
- **A. Senapati**, T. Sardar, J. Chattopadhyay. (2019). "A cholera metapopulation model interlinking migration with intervention strategies - A case study of Zimbabwe (2008-09)". *Journal of Biological Systems*, 27: 185-223. doi:10.1142/S0218339019500098.

^{*} Corresponding author, ¹ Equal contribution

SCHOLARSHIPS, FELLOWSHIPS & GRANTS

- **CDA short term fellowship 2026**, Research visit to Mahidol-Oxford Tropical Medicine Research Unit (MORU), Thailand, 2026
- **ECDC Observership 2026**, The European Centre for Disease Prevention and Control (ECDC), Solna, Swede, 2026
- **WHO Travel grant**, Global Vaccine and Immunization Research Forum (GVIRF), Brazil, 2025.
- **Travel Fellowship**, Centre for Co-operation in Science and Technology among Developing Societies (CCSTDS), India, 2019.
- **International Travel Grant**, Council of Scientific & Industrial Research (CSIR), India, 2019.
- **Senior Research Fellowship**, Council of Scientific & Industrial Research (CSIR), India, 2017.
- **Junior Research Fellowship**, Council of Scientific & Industrial Research (CSIR), India, 2014.
- **INSPIRE Scholarship**, India, 2010-2014.

AWARDS

- **People's award for best Poster at MIDSEA Summer School 2024**, Quy Nhon, Vietnam.
- **Oral presentation award – 2nd Prize at International Conference 5th India Biodiversity Meet 2018**, Indian Statistical Institute, India.

PRESENTATIONS

Invited

- “Modelling dengue vaccination in Singapore” at **Webinar Series on Health Research** organized by **Euraxess**, 2026.
- “Modelling dengue vaccination strategies in Singapore” at **Dengue and Other Arboviruses webinar series** organized by **WHO Collaboratory**, 2026.
- “Modelling the impact of dengue vaccination in Singapore” at **Indonesia-UK Vector-borne disease modelling workshop, Indonesia**, 2026.
- “Modelling dengue vaccination at Singapore” at **CDA-SSHSPH Research Forum**, Singapore, 2026.
- “Modelling dengue vaccination in Singapore” at **First Dengue Modelling Consortium Stakeholder meeting**, Singapore, 2026.
- “Mathematical modelling of dengue transmission and vaccination strategies” at **Foreign Guest Lecture** organized by Vellore Institute of technology, India, 2026.
- “Modelling impact of extreme weather events on dengue transmission” at **Virtual Short Talk Series on Climate and Health Modelling** organized by MIDSEA, 2026.

Contributed

- “Quantifying the impact of vaccination on dengue dynamics in Singapore” at **Infectious Disease Modelling Conference**, Thailand, 2024.
- “Quantifying the impact of cross-border flux on disease dynamics” at **Models in Population Dynamics, Ecology and Evolution**, Italy, 2022.
- “Impact of adult mosquito control on dengue prevalence: A case study in Kolkata” at **11th Conference on Dynamical Systems Applied to Biology and Natural Sciences**, Italy, 2020.
- “A cholera metapopulation model for estimating the impact of intervention: A case study in Zimbabwe” at **International Conference 5th India Biodiversity Meet**, India, 2018.
- “Optimal control and cost-effectiveness analysis of cholera intervention strategies in a multi-patch setting” at **National Conference on Mathematical and Theoretical Biology**, India, 2017.

Poster

- “Modelling dengue vaccination with TAK-003 in Singapore” at the launch of **The Centre for Epidemic Research and Modelling (CERM)**, Singapore, 2026.
- “Estimating the impact of age-targeted dengue vaccination Singapore” at **International Conference on Modelling and Informatics for Public Health**, India, 2025.
- “Assessing population impact of dengue vaccination strategies” at **Global Vaccine and Immunization Research Forum**, Brazil, 2025.
- “Modelling dengue vaccination strategies in Singapore” at **MIDSEA Summer School**, Vietnam, 2024.
- “Modelling framework for estimating the impact of dengue vaccination programs” at **Epidemics 9**, Italy, 2023.
- “Modelling cross-border mobility during disease outbreak” at **12th European Conference on Mathematical and Theoretical Biology**, Germany, 2022.

LECTURE, TRAINING, & MENTORING

- **Contributed** to organizing the launching of The Centre for Epidemic Research and Modelling (**CERM**), SSHSPH, NUS, February 2026.
- **Mentoring** (with the support of P.I A/Prof. Hannah E. Clapham) research intern Keya Chintan Shah under NSWS, working on the project “Modelling dengue transmission and vaccinations among migrant workers in Singapore” (December 2024 - present).
- **Invited lecture** on **Modelling impact of intervention** (course **sph3107**) at SSHSPH, NUS, Singapore, January 2025 and February 2026.
- **Hands-On Training Program** on **Simulations with MATLAB**, JIS College of Engineering, India, September 2021.

PROFESSIONAL CONTRIBUTIONS

- **Contributing** technical expertise to the development of WHO-supported Dengue simulation

platform, **Dengue Simulator** (2026 - present).

- **Reviewer** for PLoS Medicine, PLoS Neglected tropical Diseases, Epidemics, Infectious diseases of Poverty, GeoHealth, Therapeutic Advances in Vaccines and Immunotherapy, International Journal of Biomathematics, Physica A, SN computer Science, Mathematics, Fractals and Fractional.
- **Member** of Dengue and Other Arbovirus CoP by **WHO Hub for Pandemic and Epidemic Intelligence**
- **Member of** Modelling Infectious Diseases in South East Asia (**MIDSEA**) network.
- **Member and Regular reviewer** for Models of Infectious Disease Agent Study (**MIDAS**) network.
- **Invited to deliver feedback in** Stakeholder Engagement Meeting – Dengue Vaccine & Wolbachia-Based Intervention, 2025, HITAP, Bangkok, Thailand.
- **ESMID Young Scientist Member (2026 -)**